

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 33)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

[4 marks]

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01 . 2

Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

[4 marks]

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01 . 3

Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

[2 marks]

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0 2 . 1 Explain how each of the following reduces the environmental impact of fishing.
[8 marks]

Square-mesh escape panels

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Minimum and maximum catchable sizes

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No-take zones

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Closed seasons

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0 2 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.
[2 marks]

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10

03 . 1

Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

03 . 2

Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

03 . 3

Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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04 . 1

Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

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04 . 2

Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

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04 . 3

Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

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0 5 . 1

Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

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0 5 . 2

Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

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0 5 . 3

Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

10

Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / m ³ ha ⁻¹ yr ⁻¹	4	16
Bird species	38	9
Ground flora species	60	4

Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

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0 6 . 2 Explain why commercial plantations are felled before the trees reach full size.
[2 marks]

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0 6 . 3 Explain why there is little demand today for coppiced or pollarded wood.
[2 marks]

10

0 7 Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 7 . 1 Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.
[3 marks]

07 . 2

Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

07 . 3

Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

10

0 8

A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

0 8**1**

Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

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0 8**2**

Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

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0 8**3**

The test gave $p = 0.02$. Explain what this means.

[2 marks]

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0 8**4**

Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

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08 . 5

Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

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09 . 1

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

- A Environmental DNA (eDNA) analysis
- B Radio tracking
- C Lincoln index (mark-release-recapture)
- D Leg rings

☐☐☐☐

09 . 2

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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09 . 3

Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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5

Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

ESP/7447/2/SET33

Extra space

10 . 2

State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

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Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS